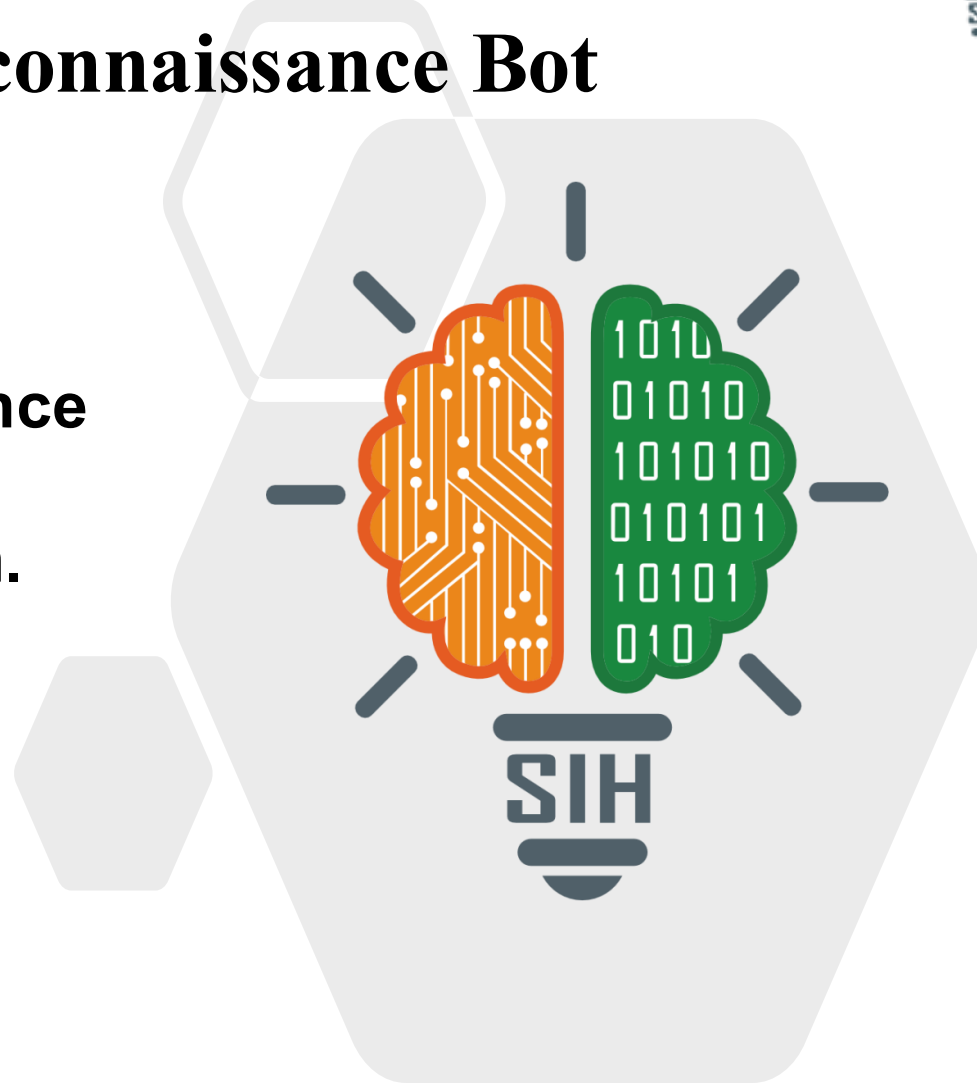


SMART INDIA HACKATHON 2025



ORB- Omni Reconnaissance Bot

- Problem Statement ID –SIH25196
- ProblemStatement Title- Surveillance
During Extreme Weather Condition.
- Theme- Miscellaneous
- PS Category- Hardware
- Team ID- ABVVSIH28
- Team Name - Alitera





Proposed Solution

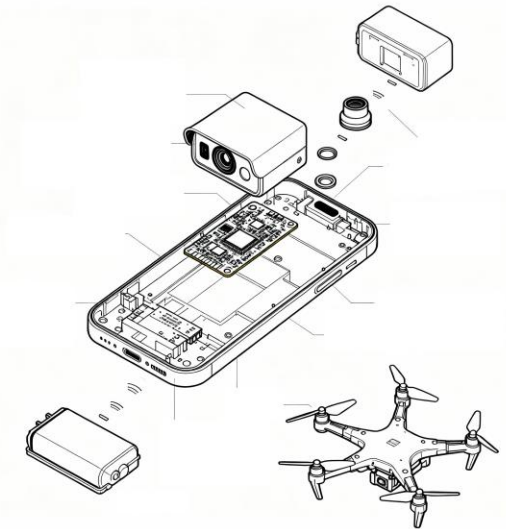
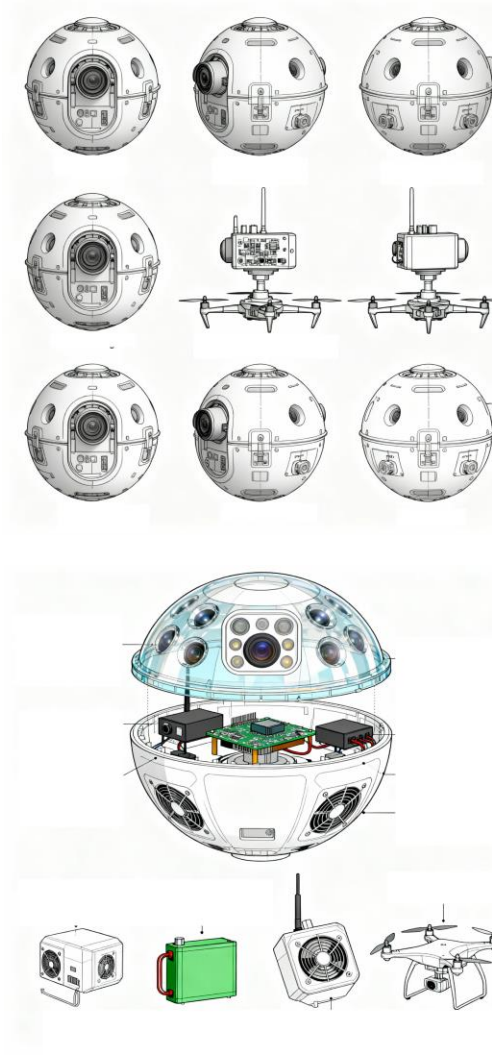
A compact palm-sized spherical device equipped with visible light cameras, thermal infrared sensors, and laser range finders for 360-degree surveillance.

A smartphone-sized surveillance tool disguised as a regular mobile device, integrating advanced multi-spectral sensors and wireless communication capabilities

The Spherical orb will be lightweight and attachable to a drone, Stick, capable of detecting human-sized targets up to 1.5 km by day and 1 km by night, even in adverse weather.

Using Principle :

<https://gensoullabs.my.canva.site/sih25196-surveillance-during-extreme-weather-condition>





TECHNICAL APPROACH



Hardware Design:

- **Orb Casing:** Hard Metallic body for Durability.
- **Sensor :** Long-Wave Infrared (LWIR) Thermal Sensor
Laser Range Finder (LRF), AM69A camera chip
Visible Light CMOS/CCD
Camera, CMOS Sensors (Mobile- Grade).

Power Supply: Use lightweight rechargeable lithium-polymer battery optimized for drone payload limits.

Transmission and Communication: Wireless, 5G

User Interface and Augmented Reality

Mounting Mechanism:

Working Concept of Device:

<https://gensoullabs.my.canva.site/sih25196-surveillance-during-extreme-weather-condition>





FEASIBILITY AND VIABILITY



Technical Feasibility:

- Key components like LWIR/MWIR sensors, compact high-resolution CMOS sensors, periscope optics, laser range finders, onboard AI processors, and wireless transmission modules are widely available and mature.
- Data Transmission: Wi-Fi, LTE/5G, and dedicated RF technologies offer real-time, low-latency video and telemetry transmission; software-defined radios can adapt to various operational environments.

Viability

Operational Benefit: Omni-directional 360° coverage with day/night, all-weather imaging dramatically enhances situational awareness for tactical and border surveillance.

Enhanced Capabilities: Integration of advanced sensors (thermal, visible, LRF) with AR HUDs improves target identification and engagement.

Cost-Effectiveness: Combining existing sensor and communication technologies reduces R&D costs and time-to-deploy.

Risks and Challenges Solutions.

- Technical challenges such as sensor calibration and environmental noise can be addressed via testing and iterative refinement and Robust Design .
- User training programs can enhance adoption and correct usage.



IMPACT AND BENEFITS



Impacts

- Enhanced situational awareness through real-time multi-sensor data access (visible, thermal, laser).
- Improved safety by enabling remote monitoring, reducing personnel exposure to threats.
- Rapid Decision Support: Instant access to fused sensor data accelerates command decisions and operational responses.
- Extended Operational Reach: Drone-mounted and portable devices provide surveillance over difficult terrain and beyond line-of-sight.
- Force Multiplication: Enables a smaller number of troops to effectively monitor larger areas, optimizing manpower use.

Benefits

- Accurate, faster target detection and classification with AI assistance.
- Cost-effective monitoring by minimizing manpower and maximizing coverage with fewer devices.
- Longer mission endurance with battery-powered operation and wireless communications.
- Enhanced decision-making quality through integrated, fused sensor data.
- Operational flexibility for various mission types including tactical, rescue, and continuous monitoring.



RESEARCH AND REFERENCES



- Fluke: What is Thermal Imaging?
<https://www.fluke.com/en-in/learn/blog/thermal-imaging/how-infrared-cameras-work>
- Accio: Laser Range Finder Module
<https://www.accio.com/plp/laser-range-finder-module>
- CircuitCentral: MPU6050 6-Axis IMU Module
<https://circuitcentral.in/products/mpu6050-6-axis-imu-module>
- DNATechIndia: Wireless Power Transfer Module
<https://www.dnatechindia.com/Wireless-Power-Transfer-Module.html>
- “Fundamentals of Sensor Technology” by Jan Fraden, Springer, ISBN: 978-3030163475
- “Embedded Systems: Introduction to ARM® Cortex™-M Microcontrollers” by Jonathan W. Valvano, ISBN: 978-1477508992



Visit the Website For More Details
About the Surveillance Device